

The Waveform-to-Bit Gap in Single-Channel Blind Source Separation: A Decision-Equivalent Evaluation Protocol

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Abstract—Separators for single-channel blind source separation (SC-BSS) of co-frequency communication signals are trained and ranked by waveform-fidelity metrics, yet the downstream task is demodulation. This letter contributes a decision-equivalent evaluation protocol—a diagnostic, oracle-compensated receiver whose hard decisions are exactly reproduced by a differentiable soft receiver—and a controlled five-seed study on a variable-count co-frequency benchmark. The protocol exposes a large waveform-to-bit gap: +4.1 dB of SI-SDR improvement moves the compensated symbol error rate from 0.5049 to 0.5027, and direct output-SIR measurement shows why: separation removes almost none of the interference (+0.3 dB SIR gain at $K=2$), the waveform gain being almost entirely noise-side. A faithfully aligned task-oriented loss moves the pooled bit error rate by at most 0.03 percentage points, but its sign is set by the number of interferers: it helps at $K=2$ and hurts at $K=3$, cancelling in the pool. A joint soft-information head fails at blind burst synchronisation. The evidence suggests closing the gap requires explicit receiver structure rather than re-weighting the tested task loss.

Index Terms—Blind source separation, co-frequency communication, evaluation protocol, symbol error rate, task-oriented learning.

I. INTRODUCTION

SINGLE-CHANNEL blind source separation (SC-BSS) of co-frequency communication signals—several modulated emitters overlapping in time and frequency at one receiving antenna—has progressed through a sequence of learned separators [1], [2], [3], [4], [5], trained and ranked by waveform-fidelity metrics, above all scale-invariant SDR (SI-SDR) [6], [7]. Yet separation is rarely the end goal: in spectrum monitoring and cognitive radio, the separated waveforms feed a demodulator, and the quantity paid for is the bit error rate (BER). The premise of the line of work is that waveform fidelity is a good proxy for bit-level recoverability. Audio research has shown such proxies can be misleading [8], [7]; for co-frequency *communication* mixtures, where the task metric is sharp (symbol and bit error rate), the premise has, to our knowledge, never been tested under statistically controlled experiments.

This letter tests it. On the variable- K co-frequency benchmark of our companion open-world study [9], we contribute:

- (i) an *offset-compensated diagnostic receiver* producing compensated symbol error rate (SER) and Gray-mapped BER from shared decisions, together with a differentiable soft-demodulation loss whose inference rule is *decision-equivalent* to it (identical argmax decisions), so the training proxy and the evaluation truth are the same receiver (Section II);
- (ii) a controlled five-seed measurement study showing the gap is real, structural, and interference-limited, with direct output-SIR evidence for the mechanism (Section IV); and
- (iii) evidence that the prescription is structural: closing the gap suggests explicit synchronisation, equalisation and joint-detection structure [10], [11], [12], rather than re-weighting the tested task loss.

II. BENCHMARK AND DECISION-EQUIVALENT PROTOCOL

The benchmark is unchanged from [9]; we restate the parameters that matter for the protocol. A single complex mixture $y(t) = \sum_{k=1}^K w_k s_k(t) + n(t)$, $t = 0, \dots, 4095$ at 16 kHz (256-symbol bursts, 16 samples/symbol, 1 kDd), of $K \in \{1, 2, 3\}$ sources drawn uniformly from {BPSK, QPSK, 8PSK, 16QAM}, root-raised-cosine pulses (roll-off 0.35, 64-tap), an independent unit-norm 3-tap complex-Gaussian fading channel per source, per-source carrier $2 \text{ kHz} + \mathcal{U}(0, 5) \text{ Hz}$ with $\pm 5 \text{ Hz}$ jitter, mixing weights $w_k \sim \mathcal{U}(0.4, 0.6)$ unnormalised, giving a pairwise source-to-source SIR of $\approx 0 \text{ dB}$ by construction, and AWGN with SNR defined on the total mixture power (per-source effective $\text{SNR} = \text{SNR} - 10 \log_{10} K$). The test grid is deterministic: 7 SNR points (-10 to 20 dB) $\times K$, 100 mixtures per cell, seed 999999.

Evaluation truth. The compensated receiver demodulates an estimate–source pair as follows: both sides are down-converted at the source’s *oracle* carrier (exposed by the generator), passed through the RRC matched filter, sampled on the zero-delay symbol grid, normalised to unit power, the estimate phase-aligned to the reference by a single least-squares complex coefficient, and decided by minimum-distance onto the constellation. Compensated SER is the fraction of mismatched hard decisions; BER is defined post-hoc via a Gray labelling of the constellation indices, from the *same* decisions. Pooled SER/BER are unweighted means over all matched estimate–source pairs. Identity input scores $\text{SER} = 0.000$ for all four modulations. We stress that this is a *diagnostic instrument*, not a *blind receiver*: by construction it factors synchronisation out of the evaluation, isolating the waveform-fidelity $\rightarrow$ symbol-decision link that is the subject of this letter.

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Decision-equivalent training proxy. The soft receiver replaces the hard decision by a cross-entropy over constellation points on the same grid,

$$\mathcal{L}_{\text{soft-SER}} = -\frac{1}{N} \sum_n \log \frac{\exp(-|z_n - c_{\ell_n}|^2/\sigma^2)}{\sum_m \exp(-|z_n - c_m|^2/\sigma^2)}, \quad (1)$$

with z_n the aligned estimate's symbols, ℓ_n the reference's hard decision, and $\sigma^2 = 0.1$ fixed on the unit-power symbol grid across all experiments (not tuned). The alignment coefficient and the labels ℓ_n are computed on detached tensors (stop-gradient), so gradients flow only through the estimated waveform. Each logit is a monotone transform of the negative squared distance, so its argmax coincides with the minimum-distance decision for any $\sigma^2 > 0$ by construction (additionally verified symbol-for-symbol on all four clean modulations): the two receivers are decision-equivalent *in their inference rule*. The soft loss is nonetheless a smooth relaxation of the hard count, so the equivalence does not extend to the optimisation dynamics — any effect it produces is attributable to the task, not to a proxy mismatch.

Statistics. All headline numbers are five-seed (42–46); the primary test is the seed-level exact sign-flip permutation (power floor $p = 0.0625$ at five seeds), with Bonferroni correction over the pre-declared contrasts. Both headline findings below were additionally replicated on three further deterministic test grids (seeds 100001–100003), so they reflect the mixture population rather than one test realisation.

III. A DECISION-LEVEL READING

The measurements below fall out of a simple observation. Write a separated estimate of a target source s as

$$\hat{s} = \alpha s + \sum_j \beta_j s_j + \nu, \quad (2)$$

with s_j the co-channel interferers and ν the noise and artefact residual. The compensated receiver normalises away the gain α and a constant phase, so its decisions depend on $\hat{s}$ only through the symbol-grid SINR $\gamma \triangleq |\alpha|^2 / (\sum_j |\beta_j|^2 + \sigma_\nu^2)$ (unit-power sources). Under a Gaussian approximation of the residual on the symbol grid, the compensated SER is a non-increasing function of γ alone: it is invariant to any reallocation of residual energy between interference and noise that preserves γ , and insensitive to reductions of σ_ν^2 while the interference term dominates. SI-SDR, in contrast, weights the two terms equally. This has two consequences.

First, a separator whose SI-SDR gain is *noise-side* leaves γ — and hence the SER — almost untouched at pairwise SIR ≈ 0 dB, where γ is saturated by the interference term. The projection-power output SIR reported below is exactly the interference factor of $1/\gamma$; it, not SI-SDR, is the decision-relevant waveform metric, which is why we measure it directly.

Second, at pairwise 0 dB the observation constellation itself overlaps: two equal-power in-phase BPSK sources coincide on the decision boundary with probability $1/2$, an irreducible $\frac{1}{4}$ error floor for that configuration that no noise suppression can lower — the collided symbol's label is simply not present in the observation. The same label-ambiguity argument bounds any decision-equivalent task loss: once this floor is reached,

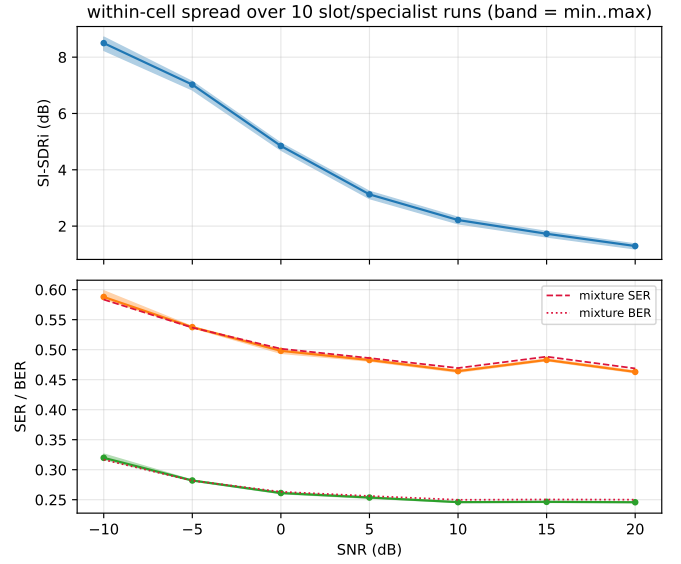


Fig. 1. The waveform-to-bit gap. Top: SI-SDR improvement over the mixture rises to +8.4 dB at −10 dB SNR (band: min–max over the ten slot/specialist runs of the honest pool; the recursive variant is excluded here because its 37% miss rate makes its matched-pair SER selection-biased). Bottom: the compensated SER/BER of the separated outputs stays glued to the mixture baseline (dashed) at every SNR — the bits do not move.

no waveform-faithful descent direction remains, which is consistent both with the sub-0.05 pp weight sweep and with the off-manifold minima the loss finds when the waveform anchor is removed (Section IV).

IV. EXPERIMENTS

SI — the mismatch is real and large. Fifteen previously trained separators — the three architectures of [9] (slot, specialist-bank, recursive one-and-rest), five training seeds each, all trained with the same MSE-anchored waveform PIT loss on the same training distribution — improve SI-SDR by +4.1 dB over the unseparated mixture on this benchmark, but move the compensated SER from 0.5049 (mixture, no separation) to only 0.5027 (BER 0.2669 → 0.2651); the per-SNR improvement of +8.4 dB at −10 dB SNR buys nothing (a fraction-of-a-point penalty, if anything) — Fig. 1 shows the whole curve. The damage localises to interference: the baseline SER jumps from 0.146 at $K=1$ to 0.561 at $K=2$ and only 0.587 at $K=3$ — one interferer closes nearly the whole demodulation gap. Direct output-SIR measurement confirms the mechanism: with $\text{SIR}(\hat{s}) = 10 \log_{10} (p_t / \sum_{j \neq t} p_j)$, where $p_j = |\langle \hat{s}, s_j \rangle|^2 / \|\hat{s}\|^2$ is the projection power of the estimate onto true source j , the per-source SIR is 0.0 dB at the mixture input and only +0.2–+0.3 dB after separation at $K=2$, and −3.1 dB at the input versus −2.9 dB after at $K=3$ — the separators suppress noise and self-distortion but pass the co-channel interferer almost untouched, and at pairwise SIR ≈ 0 dB it is the interferer, not the noise, that sets the decisions (the first consequence of Section III: the gain is noise-side, so γ and the SER do not move). A fixed- $K=2$ SIR sweep shows the flatness persists to input SIR = 20 dB (gap to the

TABLE I

PER- K EFFECT OF THE SOFT-SER TERM (FAIR ADDITIVE CONTRAST, MEAN $\pm$ STD OVER FIVE SEEDS; $\Delta\text{SER} = \text{SER}_{\text{base}} - \text{SER}_{\text{task}}$ IN PERCENTAGE POINTS, SO POSITIVE = THE TASK TERM HELPS, AND LIKEWISE FOR ΔBER). AT $K=2$ THE BER GAIN IS POSITIVE IN 5/5 SEEDS IN ALL FOUR PRE-DECLARED CONTRASTS; AT $K=3$ THE SER CHANGE IS NEGATIVE IN 5/5.

K	ΔSER (pp)	ΔBER (pp)	sign consistency
1	$+0.53 \pm 1.22$	$+0.26 \pm 0.64$	unstable (no interference)
2	$+0.32 \pm 0.35$	$+0.11 \pm 0.11$	5/5 seeds (BER, all contrasts)
3	-0.60 ± 0.13	-0.10 ± 0.10	5/5 seeds negative (SER)

per-cell baseline never exceeds 1 percentage point and sign-flips), so the interference limitation is not an artefact of the symmetric operating point. Nor is it an artefact of the test grid: on three further independent test populations the separated SER (0.504–0.508) stays glued to its mixture baseline (0.506–0.510) at the same +4.0 dB SI-SDR improvement.

S2 — a faithfully aligned task loss barely moves it, and the sign is set by the interference. Pooled over K , the soft-SER term’s effect on SER/BER is practically negligible (at most 0.17 pp for the replacement configuration; ≤ 0.03 pp for the fair additive comparison) and not statistically resolvable at five seeds. Decomposed by source count (Table I), however, the term *helps* at $K=2$ (BER +0.06 to +0.40 pp across all four pre-declared contrasts, 5/5 seeds positive in each, exact permutation $p = 0.0625$) and *hurts* at $K=3$ (SER -0.6 to -0.7 pp, 0/5 seeds positive) — the two cancelling in the pool. The ordering tracks the residual interference level (output SIR +0.3 dB at $K=2$ versus -2.9 dB at $K=3$), although five seeds cannot resolve the mechanism; it replicates on all three additional test grids ($K=2$ BER positive in 14/15 seed-grid cells, $K=3$ SER negative in 15/15). Nor is the null a matter of the task weight: sweeping $\lambda_{\text{soft-SER}}$ over three orders of magnitude (0.01–10, three seeds each) moves the pooled BER by less than 0.05 pp for every weight that preserves waveform quality, while $\lambda=10$ only degrades both objectives (SI-SDRi $+4.1 \rightarrow -0.5$ dB, pooled SER +0.6 pp). Deprived of the waveform anchor, the same loss is exploited by the optimiser (SI-SDR improvement -9.4 dB, SER *worse* than the unseparated mixture): the proxy admits off-manifold minima. With the anchor, the term instead repairs the emergent occupancy gating (occupancy-routed count accuracy $0.65 \rightarrow 0.85$; the tol-1 accuracy is 0.97–1.00 throughout, so this is threshold recalibration, not new counting ability).

S3 — naive joint detection fails, informatively. S1 answers question (A) — are the separated waveforms demodulable at all — by construction; we then ask (B): can symbol-level soft information be learned *directly* from the separated representations? A learned per-symbol soft-information head over the separated slots sits near chance ($K=2$ SER 0.732, vs. 0.561 for the mixture baseline and 0.548 for the waveform route). Controlled probes (same head, same data, oracle vs. blind synchronisation) attribute the failure to burst-level blind carrier/phase recovery: at $K=1$ an oracle-synced head merely ties the baseline (0.150 vs. 0.146), and at $K=2$ even a per-source frequency oracle fails (0.589, worse than the baseline). The gap is therefore not reducible to a synchronisation

problem: sync alone is insufficient under phase entanglement, and what remains is a separation–synchronisation–detection coupling.

V. CONCLUSION

On a co-frequency SC-BSS benchmark, waveform-fidelity gains do not reach the bits: direct output-SIR measurement shows the residual on any separated output is dominated by amplitude-level interference leakage at pairwise SIR ≈ 0 dB, and a decision-equivalent task loss shifts decisions by at most a fraction of a point, with its sign fixed by the number of interferers. Within the tested benchmark, protocol and task-loss family — including a three-orders-of-magnitude sweep of the task weight — the prescription this *suggests* is structural: synchronisation, equalisation and joint detection built into the pipeline; whether alternative task-loss families or proxies can recover part of the gap remains open. The protocol, loss and logged study are released for reuse.

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